Department of Community Medicine
Government Medical College, Akola
II/II M.B.B.S.
2022 Regular Batch (CBME)
2nd Internal Assessment Examination

Date: 22-03-2024 **Time: Marks: 50** **Total duration - 2 hrs**

Instructions:

*1) Use **blue/black** ball point pen only.*
*2) **Do not** write anything on the **blank portion of the question paper**. If written anything, such type of act will be considered as an attempt to resort to unfair means.*
*3) All questions are **compulsory**.*
*4) The number to the **right** indicates **full** marks.*
*5) Draw diagrams **wherever** necessary.*
6) Distribution of syllabus in Question Paper is only meant to cover entire syllabus within the stipulated frame. The Question paper pattern is a mere guideline. Questions can be asked from any paper's syllabus into any question paper. Students cannot claim that the Question is out of syllabus. As It is only for the placement sake, the distribution has been done.
7) Use a common answerbook for all sections.

2. Short Answer Questions (Answer Any 4 out of 5) (7x4=28 Marks)

a) In a village large number of children suffering from fever were reported from particular area. After community survey large number of mosquito larvae were found in flower vases and overhead tank in that area. Which mosquito larvae might be present in the given water source? What are different integrated mosquito control measures? (1+6 Marks)

b) Name the air pollutants emitted from automobiles. What are various preventive and control measures of air pollution? (1+6 Marks)

c) Principles of chlorination. (7 Marks)

d) In a house of 2 rooms 4 persons were living. Comment whether overcrowding present in this house or not. Describe the relationship between housing and health. (1+6 Marks)

e) Advantages and disadvantages of Natural Insecticides (4+3 Marks)

Q.3. Long Answer Questions (Answer Any 4 out of 5) (12x1=12 Marks)

Define safe and wholesome water. Enumerate various sources of water. Classify methods of purification of water. Compare Slow sand and Rapid sand filters. (1+2+3+6 Marks)

		pertain to consent for surgical procedures.
Microbiology	Module 2.2 A	Describe and discuss the role of non-malfeasance as a guiding principle in patient care
	Module 2.2 B	Describe and discuss the role of autonomy and shared responsibility as a guiding principle in patient care

Anticoagulants

An anticoagulant is a substance that prevents coagulation (clotting) of blood..

1) A group of pharmaceuticals called anticoagulants can be used *in vivo* as a medication for thrombotic disorders.
2) Anticoagulants are used in medical equipment, such as test tubes, blood transfusion bags, and renal dialysis equipment.

Laboratory instruments, blood transfusion bags, and medical and surgical equipment will get clogged up and become nonoperational if blood is allowed to clot. In addition, test tubes used for laboratory blood tests will have chemicals added to stop blood clotting.

ETHYLENEDIAMINE TETRA ACETIC ACID.(EDTA)

PRINCIPLE : EDTA is a polyprotic acid containing four carboxylic acid groups and two amine groups with lone-pair electrons that chelate calcium and several other metal ions. Calcium is necessary for a wide range of enzyme reactions of the coagulation cascade and its removal irreversibly prevents blood clotting within the collection tube. It is in a powdered (anhydrous) form .Recommended as anticoagulant because of excellent preservation of cell components and morphology of cells. In vitro diagnostics, anticoagulants are commonly added to collection tubes either to maintain blood in the fluid state for hematological testing or clinical chemistry analyses.

It uses are

- For routine hemotological test(CBC, platelet count, retic count,)
- Recommended conc. (By ISH) is dipotassium salt at a conc of 1.5+0.25 mg/ml of blood
- Tripotassium salts produce shrinkage of red cells which result in 2-3% decrease in PCV. There are negligible changes are seen in dipotassium salt when used .
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LIMITATIONS

- Excess EDTA(> 2 mg/dl) causes shrinkage and degenerative changes in red cells and leukocytes.
- Significant decrease in PCV by centrifugation and increase in MCHC
- Platelets swells then disintegrate and causes false high platelet count
- Blood film made from EDTA may fail to demonstrate basophilic strippling in lead poisoning
- Leukoagglutionation of neutrophils and lymphocytes may be seen.